



Data-Driven Farming

Harnessing the Power of AI and Machine Learning in Agriculture

Edited by

Dr. Syed Nisar Hussain Bukhari

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Data-Driven Farming

In the dynamic realm of agriculture, artificial intelligence (AI) and machine learning (ML) emerge as catalysts for unprecedented transformation and growth. The emergence of big data, Internet of Things (IoT) sensors, and advanced analytics has opened up new possibilities for farmers to collect and analyze data in real-time, make informed decisions, and increase efficiency. AI and ML are key enablers of data-driven farming, allowing farmers to use algorithms and predictive models to gain insights into crop health, soil quality, weather patterns, and more. Agriculture is an industry that is deeply rooted in tradition, but the landscape is rapidly changing with the emergence of new technologies.

Data-Driven Farming: Harnessing the Power of AI and Machine Learning in Agriculture is a comprehensive guide that explores how the latest advances in technology can help farmers make better decisions and maximize yields. It offers a detailed overview of the intersection of data, AI, and ML in agriculture and offers real-world examples and case studies that demonstrate how these tools can help farmers improve efficiency, reduce waste, and increase profitability. Exploring how AI and ML can be used to achieve sustainable and profitable farming practices, the book provides an introduction to the basics of data-driven farming, including an overview of the key concepts, tools, and technologies. It also discusses the challenges and opportunities facing farmers in today's data-driven landscape. Covering such topics as crop monitoring, weather forecasting, pest management, and soil health management, the book focuses on analyzing data, predicting outcomes, and optimizing decision-making in a range of agricultural contexts.



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*This book is dedicated to my loving father, the late Syed Gulam Mohi ud
Din Bukhari (Aba Jan) and my loving and caring mother, the late Zareefa
Syedah, whose wisdom and love continue to inspire my quest for
knowledge and innovation.*



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Editor Biography

Dr. Syed Nisar Hussain Bukhari is an accomplished professional with a strong background in academia, research, and administration. He is currently serving as Scientist-D at NIELIT Srinagar and has made significant contributions to the field of information technology (IT).

Dr. Bukhari obtained his bachelor's and master's degrees in computer applications from the University of Kashmir, Srinagar, India, and PhD in the area of machine learning, bioinformatics, and immunoinformatics from the University Institute of Computing, Chandigarh University, India.

Dr. Bukhari's research interests encompass various cutting-edge fields, including machine learning and deep learning, bioinformatics, and immunoinformatics. His work has been recognized in prestigious SCI journals such as *Nature Scientific Reports*, *Springer*, *MDPI*, and *Hindawi*, reflecting the significance and impact of his contributions to the scientific community.

Dr. Bukhari's commitment to imparting knowledge is evident through his extensive teaching experience. He has taught various core subjects, including machine learning, Python, data structures, web programming using ASP.Net, and more to postgraduates – namely, MCA and MSc IT. His comprehensive knowledge and skills have allowed him to impart training and internships in artificial intelligence, machine learning, deep learning, and other emerging technologies to engineering students.

With a passion for knowledge dissemination, Dr. Bukhari has published numerous papers in reputed journals and conferences. Dr. Bukhari's academic achievements extend beyond research publications. He has edited and authored books and received recognition as a certified machine learning expert from Stanford University and Johns Hopkins University. Additionally, he has been invited as a speaker to various seminars, webinars, and conferences, showcasing his expertise in the field. His commitment to excellence has been acknowledged with the receipt of the Best Paper Awards at multiple international conferences.

His expertise has been sought by prestigious journals like *Briefings in Bioinformatics* by Oxford University Press, *Computers in Biology and*

Medicine by Elsevier, and *Scientific Reports* by Nature, where he serves as an invited reviewer.

As a session chair, he has chaired various special sessions in Scopus-indexed international conferences. His research interests include artificial intelligence and machine learning, deep learning, bioinformatics, immunoinformatics, computational biology, and web security.

In his role as the Head of the Department of IT at NIELIT Srinagar, Dr. Bukhari demonstrated exceptional leadership and played a vital role in the growth and development of the department. During his tenure, he played a pivotal role in the department's growth and development, overseeing the MSc IT and BSc IT courses. He has provided consultancy services to different departments of the Government of Jammu and Kashmir, India, offering his expertise in implementing various e-governance-related projects.

Dr. Bukhari holds memberships with the Institution of Electronics and Telecommunication Engineers (IETE) and the International Association of Engineers (IAENG).

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Preface

It is with great pleasure and enthusiasm that I present to you *Data-Driven Farming: Harnessing the Power of AI and Machine Learning in Agriculture*. This book stands as a testament to the dynamic intersection of technology and agriculture, an evolution that has significantly transformed the way we approach farming practices. In recent years, the integration of artificial intelligence (AI) and machine learning (ML) into the agricultural landscape has marked a new era—a progressive shift from traditional methods to more sophisticated, efficient, and sustainable approaches.

The genesis of this book stems from a growing recognition of the pivotal role technology plays in addressing the complex challenges faced by the agricultural industry. The need for increased food production to meet the demands of a burgeoning global population, coupled with the imperative to manage resources more sustainably, has spurred a revolution. The utilization of AI and ML presents a compelling solution, enabling farmers, researchers, and stakeholders to optimize production, streamline processes, and make informed decisions crucial for the future of agriculture.

This book is a collaborative effort, encompassing the expertise and insights of various contributors from diverse fields—agricultural science, data science, technology, and beyond. Their collective wisdom and experiences have been meticulously crafted into a comprehensive guide that explores the multifaceted applications of AI and ML in agriculture. Readers will discover a wealth of knowledge, practical applications, and case studies that illustrate the transformative impact of data-driven approaches in farming.

The chapters contained within this book offer an in-depth exploration of the symbiotic relationship between AI, ML, and agriculture. From leveraging the Internet of Things (IoT) for precision health monitoring in livestock to the deployment of deep learning for crop disease detection, the depth and breadth of knowledge shared by our contributors demonstrate the versatility and promising implications of these technologies in agriculture.

This book is intended for a diverse readership—academics, researchers, farmers, technology enthusiasts, policymakers, and anyone interested in the transformative power of technology in agriculture. I hope that the

insights presented within these pages will serve as a valuable resource, igniting curiosity, fostering innovation, and guiding the evolution of agriculture into a more sustainable and efficient future.

I extend my heartfelt gratitude to all the contributors whose dedication and expertise have made this book a reality. I also express my appreciation to CRC Press, Taylor & Francis Group, for their support and commitment to disseminating knowledge in this ever-evolving field.

**Warm Regards,
Dr. Syed Nisar Hussain Bukhari
Editor**

Acknowledgments

The compilation of *Data-Driven Farming: Harnessing the Power of AI and Machine Learning in Agriculture* has been a journey filled with collaboration, support, and dedication from a multitude of individuals and organizations. The completion of this book would not have been possible without the invaluable contributions and unwavering support from numerous places.

First and foremost, I would like to submit all praises and thanks to **Almighty Allah**, the most beneficent and merciful, who blessed me with strength, protection, and patience to overcome all odds and made this work to come to fruition.

I extend my sincere gratitude to all the contributors, whose expertise, dedication, and passion have shaped this book. Each chapter is a testament to their commitment to advancing the knowledge frontier in the dynamic field of data-driven agriculture. Their insights, research, and real-world experiences have been instrumental in crafting a comprehensive resource for the industry, academia, and all stakeholders interested in the intersection of technology and agriculture.

I would also like to express my deepest appreciation to CRC Press, Taylor & Francis Group, for their continuous support, guidance, and belief in this project. Their commitment to disseminating knowledge and fostering the publication of innovative works has been pivotal in bringing this book to fruition.

Additionally, I am profoundly grateful to the reviewers and editors who dedicated their time and expertise to ensuring the academic rigor and quality of this publication. Their insightful feedback and meticulous review have undoubtedly enhanced the overall standard of the content, enriching the readers' experience and the credibility of the book. The support and encouragement from academic and research institutions have played a crucial role in shaping this book. I express my appreciation to all the institutions that have fostered an environment conducive to research, innovation, and the exploration of cutting-edge technologies in agriculture.

I extend my heartfelt thanks to the farmers, agricultural experts, and industry professionals whose practical insights and experiences have been

the backbone of this book. Their hands-on knowledge and willingness to adopt new technologies have inspired and guided the direction of this publication, ensuring its relevance and practicality in real-world agricultural settings.

I also express my gratitude to the readers of this book. I sincerely hope that the information and insights contained within these pages serve as a valuable resource, igniting curiosity, inspiring innovation, and fostering a deeper understanding of the powerful applications of AI and ML in agriculture.

Last but not least, I am deeply grateful for the unwavering support and inspiration provided by my late father and mother. Their enduring influence, values, and encouragement continue to resonate in every aspect of my life and work. Their belief in the pursuit of knowledge, their unwavering support during my endeavors, and their enduring love have been the guiding force behind this work. I carry their teachings of resilience, integrity, and the value of continuous learning as guiding principles that have shaped this project. Thank you to everyone who has contributed, supported, and been part of this endeavor. Your collective efforts have been instrumental in the creation of this comprehensive resource, and I am deeply appreciative of your involvement in this significant project.

Warm Regards,
Dr. Syed Nisar Hussain Bukhari
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